

Functions & Graphs

ANSWER KEY



Function transformations

- 1 Let $f(x) = x^2$ and $g(x) = -3f(x+2) - 4 = -3(x+2)^2 - 4$. Find $g(0)$, and describe the transformation from f to g in words.

$g(0) = -3(2)^2 - 4 = -3(4) - 4 = -16$. g is f shifted left 2, reflected over the x -axis and stretched vertically by a factor of 3, then shifted down 4.

- 2 Using the same $g(x) = -3(x+2)^2 - 4$, find $g(-2)$, and explain why this is the vertex of g 's graph.

$g(-2) = -3(0)^2 - 4 = -4$. Since $f(x) = x^2$ has its vertex (minimum) at $x = 0$, and g shifts f left by 2, the transformed vertex sits at $x = -2$.

Even and odd functions

- 3 Classify $f(x) = x^4 + 2x^2 - 5$ as even, odd, or neither.

Only even-degree terms appear (x^4 , x^2 , and the constant), so by Theorem 3.1 f is even. Check: $f(2) = 16 + 8 - 5 = 19$ and $f(-2) = 16 + 8 - 5 = 19$ ✓.

- 4 Classify $h(x) = 2x^5 - 3x$ as even, odd, or neither.

Only odd-degree terms appear (x^5 , x), so by Theorem 3.1 h is odd. Check: $h(1) = 2 - 3 = -1$ and $h(-1) = -2 + 3 = 1 = -h(1)$ ✓.

- 5 Explain why $k(x) = x^2 + x$ is neither even nor odd.

k mixes an even-degree term (x^2) with an odd-degree term (x), so Theorem 3.1 doesn't apply to the whole function. Checking directly: $k(1) = 2$ and $k(-1) = 0$, which is neither equal to $k(1)$ (even) nor $-k(1) = -2$ (odd).

Domain restrictions

- 6 Find the domain of $f(x) = \sqrt{2x+6}$.

Need $2x+6 \geq 0 \Rightarrow x \geq -3$. Domain: $x \geq -3$.

- 7 Find the domain of $g(x) = \frac{1}{x^2-25}$.

Need $x^2 - 25 \neq 0 \Rightarrow x \neq \pm 5$. Domain: all reals except 5 and -5.

Piecewise functions

8 Let $p(x) = 2x + 3$ for $x < 1$, and $p(x) = x^2 - 1$ for $x \geq 1$. Find $p(-2)$ and $p(3)$.

Since $-2 < 1$, use the first piece: $p(-2) = 2(-2) + 3 = -1$. Since $3 \geq 1$, use the second piece:
 $p(3) = 3^2 - 1 = 8$.